

7.2 Analysis of Windowing Effects on Spectral Estimation Using Rectangular, Hamming, and Hanning Windows in MATLAB

Code:

```
clc;
```

```
clear;
```

```
close all;
```

```
% Signal
```

```
fs = 1000;
```

```
N = 256;
```

```
t = (0:N-1)/fs;
```

```
x = sin(2*pi*100*t) + 0.8*sin(2*pi*110*t);
```

```
NFFT = 2048;
```

```
f = (0:NFFT/2-1)*(fs/NFFT);
```

```
% Windows
```

```
w = {rectwin(N), hamming(N), hann(N)};
```

```
names = {'Rectangular','Hamming','Hanning'};
```

```
% Processing
```

```
for i = 1:3
```

```
    xw = x(:).*w{i};
```

```

X = fft(xw, NFFT);

P{i} = abs(X(1:NFFT/2));
P{i} = P{i}/max(P{i});
end

%% Individual Spectrums
figure;

for i = 1:3
    subplot(3,1,i);
    plot(f,20*log10(P{i}),'LineWidth',1.5);
    title(['Spectrum using ' names{i} ' Window']);
    xlabel('Frequency (Hz)');
    ylabel('Magnitude (dB)');
    xlim([0 250]);
    grid on;
end

%% Comparison
figure;

plot(f,20*log10(P{1}),'k', ...
     f,20*log10(P{2}),'r', ...
     f,20*log10(P{3}),'b','LineWidth',1.5);

```

```

legend(names);
title('Window Comparison');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
xlim([50 170]);
grid on;

%% Window Shapes
figure;

for i = 1:3
    subplot(3,1,i);
    plot(w{i},'LineWidth',1.5);
    title([names{i} ' Window']);
    grid on;
end

%% Peak Detection
for i = 1:3
    [pks{i},loc{i}] = findpeaks(P{i},f,'MinPeakHeight',0.3);

    fprintf('\n%s Window Frequencies:\n',names{i});
    disp(loc{i});
end

```

```

%% Peak Marking (Hamming)

figure;

plot(f,20*log10(P{2}),'LineWidth',1.5);

hold on;

plot(loc{2},20*log10(pks{2}),'ro','MarkerFaceColor','r');

title('Detected Frequencies (Hamming)');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
xlim([50 170]);
grid on;

%% Main Peaks

fprintf('\nMain Spectral Peaks:\n');

for i = 1:3

    [~,idx] = max(P{i});

    fprintf('%s Window Peak = %.2f Hz\n', ...
        names{i},f(idx));

end

Output:

```



